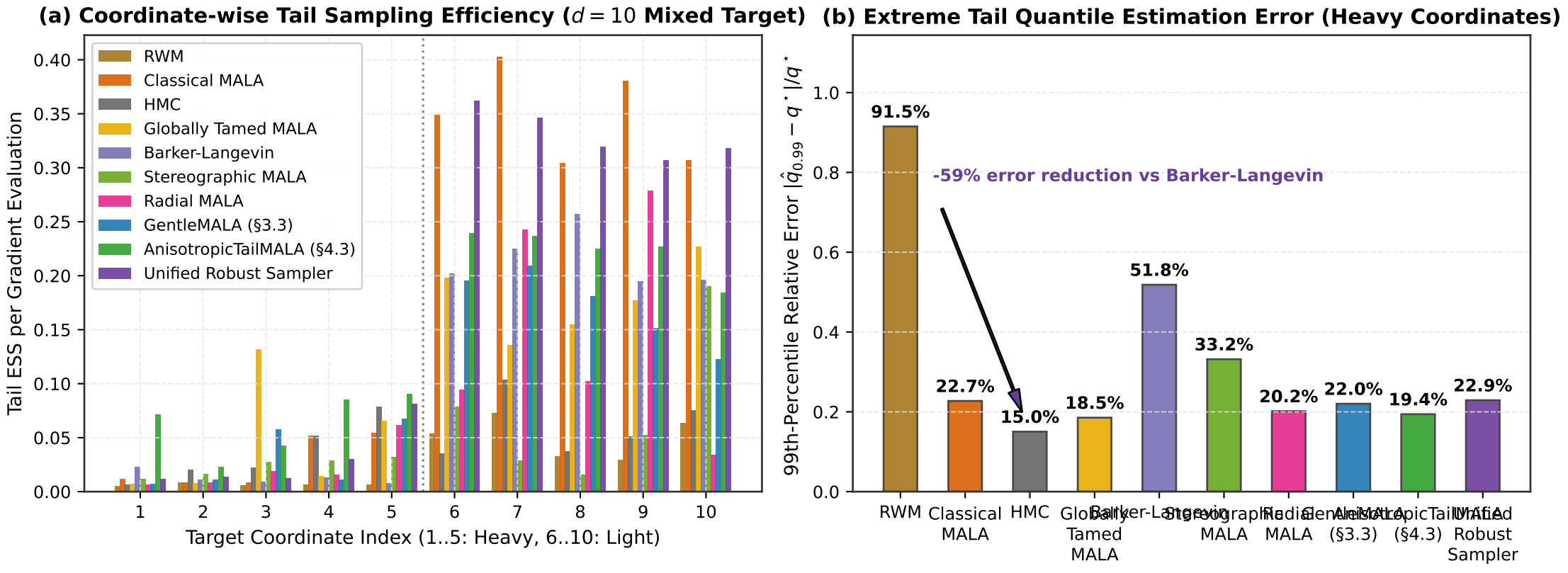
(a) Coordinate-wise Tail Sampling Efficiency ($d = 10$ Mixed Target)
Tail ESS per Gradient Evaluation
Target Coordinate Index (1..5: Heavy, 6..10: Light)
RWM
Classical MALA
HMC
Globally Tamed MALA
Barker-Langevin
Stereographic MALA
Radial MALA
GentleMALA (§3.3)
AnisotropicTailMALA (§4.3)
Unified Robust Sampler
(b) Extreme Tail Quantile Estimation Error (Heavy Coordinates)
99th-Percentile Relative Error $|\hat{q}_{0.99} - q^\star|/q^\star$
91.5%
22.7%
15.0%
18.5%
51.8%
33.2%
20.2%
22.0%
19.4%
22.9%
-59% error reduction vs Barker-Langevin
RWM
Classical MALA
HMC
Globally Tamed MALA
Barker-Langevin
Stereographic MALA
Radial MALA
GentleMALA (§3.3)
AnisotropicTailMALA (§4.3)
Unified Robust Sampler